

Therapeutic Equipment

Complete student lesson for course code **4242**.

Revision 2026 Semester 4 Program Core 4 modules

Course snapshot

Scheme: 4 (L:3,T:1,P:0); 60 periods

Credits: 4

Contact: 4 (L:3,T:1,P:0)

Module hours listed: 43

Official Source Declaration

This lesson uses the supplied official syllabus PDF as the authoritative source for course information, revision, modules, topics, objectives, Course Outcomes, assessment information, official activities, practical requirements, and references.

Source handling: Official wording and numerical values are retained wherever supplied. Educational explanations, Malayalam support, examples, diagrams, and revision questions are added as study support and are not presented as additional official requirements.

Transparency note: Official assessment information is reproduced below from the supplied syllabus. The supplied PDF is titled “Therapeutic Equipment”, but its course outline covers AC circuits, network theorems, transformers, DC machines and AC machines; this apparent title/content inconsistency is retained and is not silently corrected. Any limitation in the supplied PDF is not silently corrected.

How to Study This Lesson

Recommended sequence

1. Begin with the official source declaration and course dashboard.
2. Study each module in the official order and keep the coverage table as a checklist.
3. Open every topic, understand the example or procedure, and write your own short answer.
4. Use the revision section, glossary, questions, and practice paper after completing the modules.

Study principle: Keep English technical terminology, symbols, units, and official assessment wording unchanged in examination answers. Use the Malayalam support blocks only as a bridge to understanding.

Course Dashboard

Course code

4242

Semester

4

Credits

4

Teaching scheme

4 (L:3,T:1,P:0); 60 periods

Module-hour and outcome mapping

No.	Module	Official hours	Relevant CO / level
1	AC signals and RLC circuits	11	CO1 · Applying
2	Network theorems and transformers	12	CO2 · Applying

No.	Module	Official hours	Relevant CO / level
3	DC machines	10	CO3 · Understanding
4	AC machines	10	CO4 · Understanding

Prerequisites

- Basic Engineering Mathematics principles; Magnetism.

How to study this lesson

1. Read the module introduction and learning goals.
2. Open every topic and reproduce its example, sequence, or procedure.
3. Use Malayalam support as a bridge; retain English technical terms for examinations.
4. Attempt the module questions without looking at the answers.

Objectives, Course Outcomes and CO-PO Information

Official course objectives

1. To develop the skill to diagnose and rectify the electric circuit networks and understand the operations of electrical machines.

Course Outcomes

1. Solve a given AC circuit to find various parameters.
2. Apply various network theorems for simplifying electric circuits and networks and illustrate the operations of transformers.
3. Illustrate the principles and operations of DC machines.
4. Illustrate the principles and operations of AC machines.

CO-PO mapping is reproduced from the supplied syllabus; the numerical mapping values are retained exactly where stated.

Complete Syllabus Coverage

Use this checklist to confirm that every official module topic is represented in the lesson.

Complete official topic coverage checklist

Module	Topic code	Topic	Hours
module-1	M1.01	AC signals and waveforms	1
module-1	M1.02	AC through resistance, inductance and capacitance	2
module-1	M1.03	Phasors, AC power and power factor	4
module-1	M1.04	Series and parallel RLC circuits and resonance	4
module-2	M2.01	Network theorems	3
module-2	M2.02	Applying network theorems	4
module-2	M2.03	Transformer principle and operation	3
module-2	M2.04	Transformer types, rating, EMF and losses	2
module-3	M3.01	DC generators	3
module-3	M3.02	DC motors and back EMF	2
module-3	M3.03	Starters for DC motors	3
module-3	M3.04	Comparison of DC motors	2
module-4	M4.01	Alternator principle and operation	2
module-4	M4.02	AC-motor principles and classification	2
module-4	M4.03	Stepper, universal and servo motors	3
module-4	M4.04	Single-phase and three-phase induction motors	3

Learning Roadmap

1. AC signals and RLC circuits

2. Network theorems and transformers

3. DC machines
CO3 · Understanding ·

**CO1 · Applying · 11
hours**

**CO2 · Applying · 12
hours**

10 hours

**4. AC machines
CO4 · Understanding ·
10 hours**

The roadmap follows the order of the supplied syllabus.

OFFICIAL MODULE 1

AC signals and RLC circuits

Alternating quantities vary with time and are represented through waveforms, phasors and RMS values. R, L and C produce different phase relationships and determine AC power and resonance.

Hours: 11

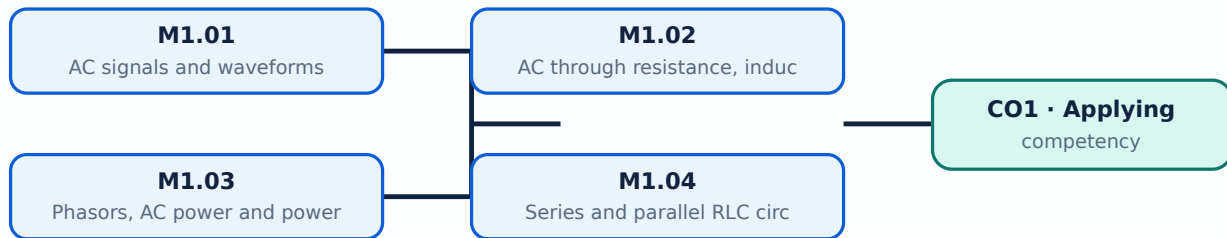
CO1 · Applying · Bloom level: Applying

Learning goals

- Define AC waveform terms.
- Analyse R-L-C behaviour.
- Calculate AC power and power factor.
- Solve simple series and parallel RLC circuits.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-1: the listed topics build the module competency.

– M1.01 — AC signals and waveforms (1 hours)

Concept and explanation

Alternating voltage and current may be sinusoidal, square, triangular or sawtooth. Cycle, time period, frequency, amplitude, phase, maximum value, RMS value, average value and form factor describe the signal.

Malayalam support: ഏകദിശീകൃതമായ വോൾട്ടേജ് കറന്റ് സൈനസ്, ചതുരശ്ര, ത്രികോണ അല്ലെങ്കിൽ സാമന്തം ആകാം. സൈക്കിൾ, സമയകാലം, ഫ്രീക്വൻസി, അംഗീകൃതം, ഫേസ്, പരമാവധി മൂല്യം, RMS മൂല്യം, ശരാശരി മൂല്യം എന്നിവ സിഗ്നൽ വിവരിക്കുന്നു.

Study points

- Define the waveform terms.
- Relate period and frequency.
- Distinguish maximum, RMS and average values.

– M1.02 — AC through resistance, inductance and capacitance (2 hours)

Concept and explanation

In a resistor, voltage and current are in phase. In an inductor, current lags voltage; in a capacitor, current leads voltage. These relationships determine impedance and phasor diagrams.

Malayalam support: ഏകദിശതരംഗത്തിലൂടെ വോൾട്ടേജ് കറന്റ് തമ്മിലുള്ള ബന്ധം. English technical terms ഫേസർ ഡയഗ്രാം.

Study points

- State phase relationships.
- Draw basic phasors.
- Identify the effect of frequency.

Engineering / workplace application: The R-L-C model is used to predict motor, filter and power-system behaviour.

– M1.03 — Phasors, AC power and power factor (4 hours)

Concept and explanation

Phasors represent sinusoidal quantities using magnitude and phase. Active power, reactive power and apparent power form the power triangle, while power factor indicates the ratio of active to apparent power.

Malayalam support: ഏകദിശതരംഗത്തിലൂടെ വോൾട്ടേജ് കറന്റ് തമ്മിലുള്ള ബന്ധം. English technical terms ഫേസർ ഡയഗ്രാം.

Study points

- Define power factor.
- Distinguish active, reactive and apparent power.
- Solve simple phasor-power problems.

Illustrative example: For a single-phase load, apparent power is $S = VI$, active power is $P = VI \cos \phi$, and reactive power is $Q = VI \sin \phi$ using RMS values.

– M1.04 — Series and parallel RLC circuits and resonance (4 hours)

Concept and explanation

Series and parallel RLC circuits have frequency-dependent impedance. Resonance occurs when inductive and capacitive effects balance; Q factor indicates the sharpness or selectivity of resonance.

Malayalam support: റ്റു സിരീസ് റ്റു പാരലൽ റ്റു സിരീസ് റ്റു പാരലൽ റ്റു സിരീസ് English technical terms റ്റു സിരീസ് റ്റു പാരലൽ.

Study points

- Solve simple RLC circuits.
- Explain series and parallel resonance.
- Define Q factor.
- Keep impedance units consistent.

Common mistake: Do not use peak voltage in an RMS power equation without converting the value first.

Module summary: Use a consistent phasor reference and keep RMS, peak and average values distinct.

OFFICIAL MODULE 2

Network theorems and transformers

Network theorems simplify electrical circuits, while transformers transfer AC power between circuits through magnetic coupling. The module combines theorem-based calculation with transformer construction and operation.

Hours: 12

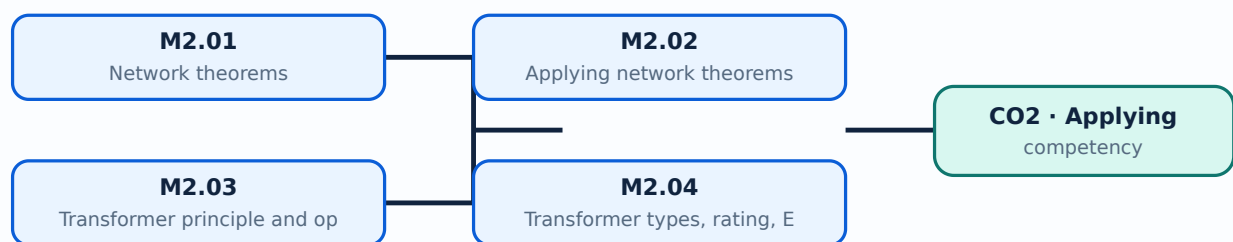
CO2 · Applying · Bloom level: Applying

Learning goals

- Apply Ohm and Kirchhoff laws.
- Use superposition, Thevenin and maximum-power-transfer theorems.
- Explain transformer construction, ratio, losses and applications.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-2: the listed topics build the module competency.

– M2.01 — Network theorems (3 hours)

Concept and explanation

Ohm's law and Kirchhoff's laws provide the basic relationships for circuit analysis. Superposition considers independent sources separately; Thevenin's theorem replaces a linear network by an equivalent voltage source and resistance; maximum power transfer identifies a load-matching condition.

Malayalam support: ഞങ്ങൾ ഇംഗ്ലീഷ് സാങ്കേതിക പദങ്ങൾക്ക് മലയാളം പദങ്ങൾ നൽകുന്നു.

Study points

- State each theorem.
- Identify a suitable theorem.
- Keep source deactivation rules correct.

– M2.02 — Applying network theorems (4 hours)

Concept and explanation

Theorems reduce a circuit to a form that makes current, voltage or power calculation easier. A solution should state the equivalent circuit, substitute values with units and verify the result.

Malayalam support: താഴെ പറയുന്നവയെക്കുറിച്ചുള്ള ആശയം ഉപയോഗിച്ച് English technical terms ഉപയോഗിച്ച് വിശദീകരിക്കുക.

Study points

- Solve a simple network problem.
- Find an equivalent source and resistance.
- Check the sign and units.

Suggested procedure

1. Define the load or required branch.
2. Remove or retain sources according to the theorem.
3. Calculate the equivalent quantities.
4. Reconnect the load and determine the required parameter.

– M2.03 — Transformer principle and operation (3 hours)

Concept and explanation

A transformer uses mutual induction between primary and secondary windings on a magnetic core. An ideal transformer changes voltage and current according to its turns ratio while conserving apparent power approximately.

Malayalam support: ഞ ഞ ഞ ഞ ഞ ഞ ഞ ഞ ഞ ഞ English technical terms ഞ ഞ ഞ ഞ ഞ ഞ ഞ.

Study points

- Explain mutual induction.
- Identify primary, secondary and core.
- State the ideal transformation ratio.

Engineering / workplace application: Transformers provide isolation and voltage conversion in power supplies and distribution systems.

– M2.04 — Transformer types, rating, EMF and losses (2 hours)

Concept and explanation

Transformer rating is expressed in apparent power. The EMF equation relates induced voltage to frequency, flux and turns. Copper, core and other losses reduce efficiency; step-up, step-down and isolation types serve different applications.

Malayalam support: ഞ ഞ ഞ ഞ ഞ ഞ ഞ ഞ ഞ ഞ English technical terms ഞ ഞ ഞ ഞ ഞ ഞ ഞ.

Study points

- List transformer types.
- State the role of rating.
- Identify principal losses and applications.

Module summary: Draw the circuit clearly, identify the required equivalent, and check the result against limiting cases.

OFFICIAL MODULE 3

DC machines

DC machines operate as generators or motors. The module covers generation of EMF, armature reaction, characteristics, back EMF, starters and comparison of motor types.

Hours: 10

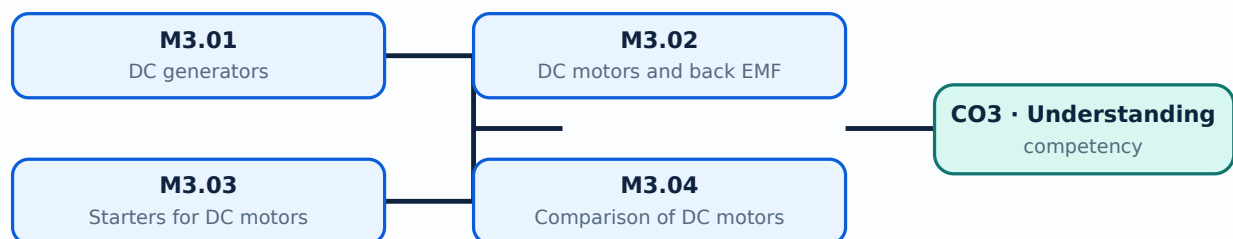
CO3 · Understanding · Bloom level: Understanding

Learning goals

- Explain DC-generator operation.
- Explain DC-motor operation and back EMF.
- Describe a three-point starter.
- Compare DC motors.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-3: the listed topics build the module competency.

– M3.01 — DC generators (3 hours)

Concept and explanation

A DC generator converts mechanical energy into electrical energy through electromagnetic induction. Types include separately excited, shunt, series and compound generators; the EMF equation and armature reaction describe important behaviour.

Malayalam support: ഡി സി ജനറേറ്റർ പ്രവർത്തനം English technical terms മെക്കാനിക്കൽ ഊർജ്ജം, വൈദ്യുത ഊർജ്ജം.

Study points

- State the generator principle.
- List generator types.
- Explain armature reaction and no-load characteristics.

– M3.02 — DC motors and back EMF (2 hours)

Concept and explanation

A DC motor converts electrical energy into mechanical energy. Back EMF opposes the applied voltage and regulates armature current as the motor accelerates.

Malayalam support: ഡി സി മോട്ടർ പ്രവർത്തനം English technical terms വൈദ്യുത ഊർജ്ജം, മെക്കാനിക്കൽ ഊർജ്ജം.

Study points

- State the motor principle.
- Explain the significance of back EMF.
- Relate current to torque conceptually.

Engineering / workplace application: DC motors are selected where speed control and high starting torque are important.

– M3.03 — Starters for DC motors (3 hours)

Concept and explanation

At starting, back EMF is initially small and armature current could become excessive. A starter inserts resistance and provides controlled acceleration; the three-point starter includes line, armature and field connections with protective features.

Malayalam support: ഏതെങ്കിലും മൂന്ന് ഇംഗ്ലീഷ് ടെക്നിക്കൽ ടേർമുകൾ തിരഞ്ഞെടുത്ത് മലയാളത്തിൽ വിശദീകരിക്കുക.

Study points

- State the necessity of a starter.
- Explain a three-point starter.
- Identify protective functions.

– M3.04 — Comparison of DC motors (2 hours)

Concept and explanation

Shunt, series and compound motors have different speed-torque characteristics. Comparison should consider starting torque, speed regulation, load behaviour and applications.

Malayalam support: ഏതെങ്കിലും മൂന്ന് ഇംഗ്ലീഷ് ടെക്നിക്കൽ ടേർമുകൾ തിരഞ്ഞെടുത്ത് മലയാളത്തിൽ വിശദീകരിക്കുക.

Study points

- Compare the motor types.
- Relate characteristics to applications.
- Use a table for an examination answer.

Module summary: A machine answer should separate construction, principle, operation, characteristic and application.

AC machines

AC machines include alternators and motors. The module introduces synchronous speed, frequency, the alternator EMF equation and the principles and applications of several AC motors.

Hours: 10

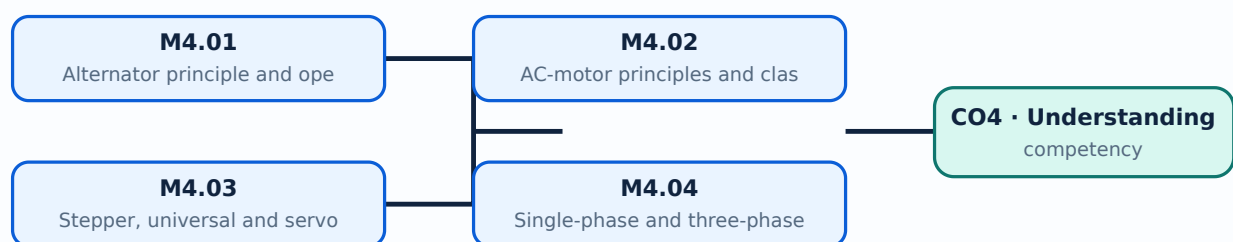
CO4 · Understanding · Bloom level: Understanding

Learning goals

- Explain alternator operation.
- Relate speed and frequency.
- Classify AC motors.
- State applications of single-phase and three-phase motors.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-4: the listed topics build the module competency.

– M4.01 — Alternator principle and operation (2 hours)

Concept and explanation

An alternator converts mechanical energy into AC electrical energy. Its EMF depends on flux, speed, frequency and winding turns; open-circuit characteristics show generated voltage behaviour.

Malayalam support: ഏകദിശീകൃത വൈദ്യുത ഊർജ്ജം മെക്കാനിക്കൽ ഊർജ്ജത്തിൽ മാറ്റം വരുത്തുന്ന യന്ത്രമാണ് ആൾട്ടറനേറ്റർ. English technical terms ഏകദിശീകൃത വൈദ്യുത ഊർജ്ജം.

Study points

- State the alternator principle.
- Explain the EMF relationship.
- Describe the open-circuit characteristic.

– M4.02 — AC-motor principles and classification (2 hours)

Concept and explanation

AC motors use alternating magnetic fields to produce torque. Classification includes induction, synchronous and special-purpose motors, each suited to different speed and control requirements.

Malayalam support: ഏകദിശീകൃത വൈദ്യുത ഊർജ്ജം ഉപയോഗിച്ച് ടോർക്ക് ഉത്പാദിപ്പിക്കുന്ന യന്ത്രമാണ് ആൾട്ടറനേറ്റർ. English technical terms ഏകദിശീകൃത വൈദ്യുത ഊർജ്ജം.

Study points

- Classify AC motors.
- State the induction-motor principle.
- Relate supply and rotating field.

– M4.03 — Stepper, universal and servo motors (3 hours)

Concept and explanation

Stepper motors rotate in discrete steps, universal motors can operate on AC or DC, and servo motors form part of a feedback-controlled position or speed system.

Malayalam support: ഞ ഞ ഞ ഞ ഞ ഞ ഞ ഞ ഞ ഞ ഞ English technical terms ഞ ഞ ഞ ഞ ഞ ഞ ഞ.

Study points

- Explain the working idea of each motor.
- List applications.
- Distinguish open-loop stepping from servo feedback.

Engineering / workplace application: Printers and positioning systems use steppers, portable tools may use universal motors, and automation uses servo motors.

– M4.04 — Single-phase and three-phase induction motors (3 hours)

Concept and explanation

Induction motors develop torque through induced rotor currents. Single-phase motors require a starting method, while three-phase motors naturally produce a rotating magnetic field and are widely used in industry.

Malayalam support: ഞ ഞ ഞ ഞ ഞ ഞ ഞ ഞ ഞ ഞ ഞ English technical terms ഞ ഞ ഞ ഞ ഞ ഞ ഞ.

Study points

- Explain the induction principle.
- Compare single- and three-phase motors.
- List applications and starting considerations.

Module summary: Use a labelled block or schematic and connect the operating principle to the application.

Formula and Notation Bank

Formula note: The supplied syllabus is primarily procedural or conceptual and does not prescribe a compulsory numerical formula bank. Focus on the official terminology, sequences, records, and practical demonstrations listed in the modules.

Diagram Library for Rapid Revision

[Open the module to view its labelled learning-map diagram.](#)

[Module 2: Network](#)

[Open the module to view its labelled learning-map diagram.](#)

[Module 4: AC machines](#)

[Open the module to view its labelled learning-map diagram.](#)

Official Activities and Practical Requirements

Official activities / self-learning

- No separate activity list is provided in the supplied PDF.

Practical requirements / equipment

- No separate practical equipment list is provided in the supplied PDF.

Assessment Methodology

Official assessment information is reproduced below from the supplied syllabus.

- Series Test: 2 hours/assessment component as stated in the supplied syllabus.
- Source transparency: The supplied PDF is titled “Therapeutic Equipment”, but its course outline covers AC circuits, network theorems, transformers, DC machines and AC machines. This apparent title/content inconsistency is retained and is not silently corrected.
- The supplied syllabus does not state a separate CIE/ESE mark distribution.

Module-wise Questions and Answers

module-1 — AC signals and RLC circuits

1. Explain AC signals and waveforms. [3 marks]

Alternating voltage and current may be sinusoidal, square, triangular or sawtooth. Cycle, time period, frequency, amplitude, phase, maximum value, RMS value, average value and form factor describe the signal.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

2. Explain AC through resistance, inductance and capacitance. [3 marks]

In a resistor, voltage and current are in phase. In an inductor, current lags voltage; in a capacitor, current leads voltage. These relationships determine impedance and phasor diagrams.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

3. Explain Phasors, AC power and power factor. [3 marks]

Phasors represent sinusoidal quantities using magnitude and phase. Active power, reactive power and apparent power form the power triangle, while power factor indicates the ratio of active to apparent power.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

4. Explain Series and parallel RLC circuits and resonance. [3 marks]

Series and parallel RLC circuits have frequency-dependent impedance. Resonance occurs when inductive and capacitive effects balance; Q factor indicates the sharpness or selectivity of resonance.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

module-2 — Network theorems and transformers

5. Explain Network theorems. [3 marks]

Ohm's law and Kirchhoff's laws provide the basic relationships for circuit analysis. Superposition considers independent sources separately; Thevenin's theorem replaces a linear network by an equivalent voltage source and resistance; maximum power transfer identifies a load-matching condition.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

6. Explain Applying network theorems. [3 marks]

Theorems reduce a circuit to a form that makes current, voltage or power calculation easier. A solution should state the equivalent circuit, substitute values with units and verify the result.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

7. Explain Transformer principle and operation. [3 marks]

A transformer uses mutual induction between primary and secondary windings on a magnetic core. An ideal transformer changes voltage and current according to its turns ratio while conserving apparent power approximately.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

8. Explain Transformer types, rating, EMF and losses. [3 marks]

Transformer rating is expressed in apparent power. The EMF equation relates induced voltage to frequency, flux and turns. Copper, core and other losses reduce efficiency; step-up, step-down and isolation types serve different applications.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

module-3 — DC machines

9. Explain DC generators. [3 marks]

A DC generator converts mechanical energy into electrical energy through electromagnetic induction. Types include separately excited, shunt, series and compound generators; the EMF equation and armature reaction describe important behaviour.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

10. Explain DC motors and back EMF. [3 marks]

A DC motor converts electrical energy into mechanical energy. Back EMF opposes the applied voltage and regulates armature current as the motor accelerates.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

11. Explain Starters for DC motors. [3 marks]

At starting, back EMF is initially small and armature current could become excessive. A starter inserts resistance and provides controlled acceleration; the three-point starter includes line, armature and field connections with protective features.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

12. Explain Comparison of DC motors. [3 marks]

Shunt, series and compound motors have different speed-torque characteristics. Comparison should consider starting torque, speed regulation, load behaviour and applications.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

module-4 — AC machines

13. Explain Alternator principle and operation. [3 marks]

An alternator converts mechanical energy into AC electrical energy. Its EMF depends on flux, speed, frequency and winding turns; open-circuit characteristics show generated voltage behaviour.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

14. Explain AC-motor principles and classification. [3 marks]

AC motors use alternating magnetic fields to produce torque. Classification includes induction, synchronous and special-purpose motors, each suited to different speed and control requirements.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

15. Explain Stepper, universal and servo motors. [3 marks]

Stepper motors rotate in discrete steps, universal motors can operate on AC or DC, and servo motors form part of a feedback-controlled position or speed system.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

16. Explain Single-phase and three-phase induction motors. [3 marks]

Induction motors develop torque through induced rotor currents. Single-phase motors require a starting method, while three-phase motors naturally produce a rotating magnetic field and are widely used in industry.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

Practice Model Question Paper

Practice Model Paper — Not an Official Examination Paper

This paper is generated from the supplied module coverage for revision and does not claim to reproduce the official examination pattern.

1. Define or explain *AC signals and waveforms* with one relevant application. (5 marks)
2. Define or explain *AC through resistance, inductance and capacitance* with one relevant application. (5 marks)
3. Define or explain *Network theorems* with one relevant application. (5 marks)
4. Define or explain *Applying network theorems* with one relevant application. (5 marks)
5. Define or explain *DC generators* with one relevant application. (5 marks)
6. Define or explain *DC motors and back EMF* with one relevant application. (5 marks)
7. Define or explain *Alternator principle and operation* with one relevant application. (5 marks)
8. Define or explain *AC-motor principles and classification* with one relevant application. (5 marks)

Writing advice: Use headings, standard terminology, and labelled diagrams or procedures where relevant.

Quick Revision

Module-wise key points

- **AC signals and RLC circuits:** Use a consistent phasor reference and keep RMS, peak and average values distinct.
- **Network theorems and transformers:** Draw the circuit clearly, identify the required equivalent, and check the result against limiting cases.
- **DC machines:** A machine answer should separate construction, principle, operation, characteristic and application.
- **AC machines:** Use a labelled block or schematic and connect the operating principle to the application.

Exam checklist

- Write the official terminology before adding explanation.
- Use a labelled diagram or step sequence when it improves the answer.
- Check conditions, boundaries, safety, hygiene, and documentation points.
- Separate official syllabus information from your own example.

Last-minute revision: Review the coverage table, formula bank, diagram library, and common-mistake notes inside every topic.

Glossary

Technical glossary

Term	Short meaning
AC signals and waveforms	Alternating voltage and current may be sinusoidal, square, triangular or sawtooth. Cycle, time period, frequency, amplitude, phase, maximum value, RMS value, average value and form factor describe the signal.
AC through resistance, inductance and capacitance	In a resistor, voltage and current are in phase. In an inductor, current lags voltage; in a capacitor, current leads voltage. These relationships determine impedance and phasor diagrams.
Phasors, AC power and power factor	Phasors represent sinusoidal quantities using magnitude and phase. Active power, reactive power and apparent power form the power triangle, while power factor indicates the ratio of active to apparent power.
Series and parallel RLC circuits and resonance	Series and parallel RLC circuits have frequency-dependent impedance. Resonance occurs when inductive and capacitive effects balance; Q factor indicates the sharpness or selectivity of resonance.

Term	Short meaning
Network theorems	Ohm's law and Kirchhoff's laws provide the basic relationships for circuit analysis. Superposition considers independent sources separately; Thevenin's theorem replaces a linear network by an equivalent voltage source and resistance; maximum power transfer theorem
Applying network theorems	Theorems reduce a circuit to a form that makes current, voltage or power calculation easier. A solution should state the equivalent circuit, substitute values with units and verify the result.
Transformer principle and operation	A transformer uses mutual induction between primary and secondary windings on a magnetic core. An ideal transformer changes voltage and current according to its turns ratio while conserving apparent power approximately.
Transformer types, rating, EMF and losses	Transformer rating is expressed in apparent power. The EMF equation relates induced voltage to frequency, flux and turns. Copper, core and other losses reduce efficiency; step-up, step-down and isolation types serve different applications.
DC generators	A DC generator converts mechanical energy into electrical energy through electromagnetic induction. Types include separately excited, shunt, series and compound generators; the EMF equation and armature reaction describe important behaviour.
DC motors and back EMF	A DC motor converts electrical energy into mechanical energy. Back EMF opposes the applied voltage and regulates armature current as the motor accelerates.
Starters for DC motors	At starting, back EMF is initially small and armature current could become excessive. A starter inserts resistance and provides controlled acceleration; the three-point starter includes line, armature and field connections with protective features.
Comparison of DC motors	Shunt, series and compound motors have different speed-torque characteristics. Comparison should consider starting torque, speed regulation, load behaviour and applications.
Alternator principle and operation	An alternator converts mechanical energy into AC electrical energy. Its EMF depends on flux, speed, frequency and winding turns; open-circuit characteristics show generated voltage behaviour.
AC-motor principles and classification	AC motors use alternating magnetic fields to produce torque. Classification includes induction, synchronous and special-purpose motors, each suited to different speed and control requirements.
Stepper, universal and servo motors	Stepper motors rotate in discrete steps, universal motors can operate on AC or DC, and servo motors form part of a feedback-controlled position or speed system.
Single-phase and three-phase induction motors	Induction motors develop torque through induced rotor currents. Single-phase motors require a starting method, while three-phase motors naturally produce a rotating magnetic field and are widely used in industry.

Official References and Resources

References listed in the supplied syllabus

1. B. L. Theraja and A. K. Theraja, Textbook of Electrical Technology: Part 1 – Basic Electrical Engineering in S.I. Units, S. Chand, 2012.
2. B. L. Theraja and A. K. Theraja, Textbook of Electrical Technology: Part 2 – AC and DC Machines, S. Chand, 2012.
3. Dr Ganesh Rao and R. Sreenivasa, Network Analysis: A Simplified Approach, Cengage.
4. J. B. Gupta, Fundamentals of Electrical Engineering and Electronics, S. K. Kataria & Sons, 2009.

Official learning resources listed

- <https://www.khanacademy.org/electrical/>
- <https://ocw.mit.edu/>
- <https://www.swayam.gov.in/>

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